



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI LAKSHMI FUEL AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

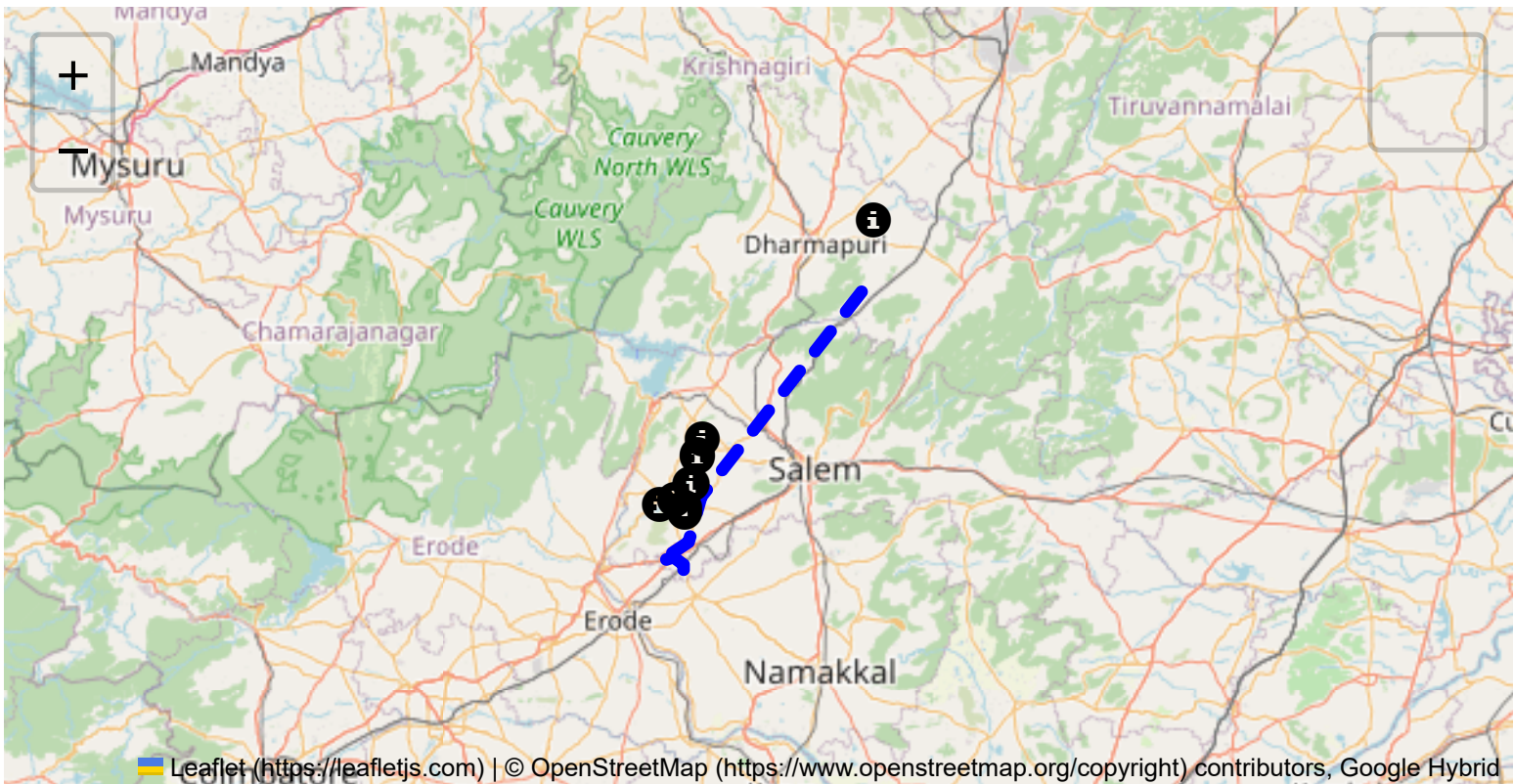
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

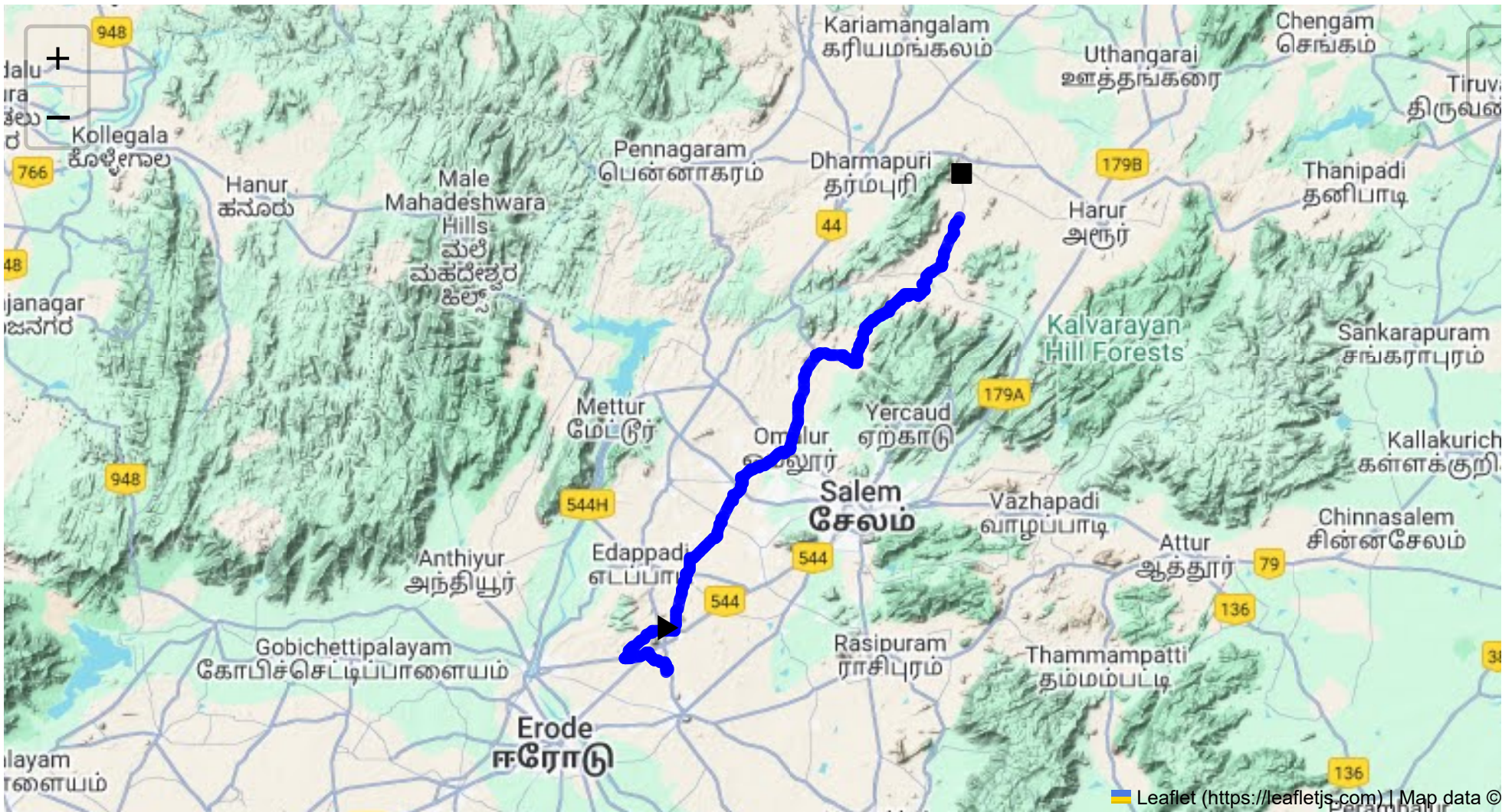
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 106.76 km
Estimated Duration: 2.6 hours
Adjusted Duration (Heavy Vehicle): 3.2 hours
Start: (11.4381, 77.8734)
End: (12.058692, 78.283821)



Welcome to the Journey Risk Management Study

Route Safety Analysis Report: Sangagiri to Dharmapuri, Tamil Nadu

1. Overview of the Route Map: The route from Sangagiri to Dharmapuri, via several key points along the SH 86 and Bhavani Main Road, covers approximately 106.76 kilometers. The journey involves traveling through both urban and rural areas in Tamil Nadu, with a mix of state highways and smaller roads. This route is predominantly rural, with some sections passing through small towns and villages.

2. Weather Conditions: Tamil Nadu typically experiences a tropical climate. During the monsoon season (October to December), heavy rains can cause flooding and road closures. High temperatures during the summer months (March to May) can lead to road surface damage and increased vehicle overheating risks. Dense fog can be an issue during early mornings and late nights, particularly in hilly areas.

3. Traffic Patterns and Congestion: Traffic tends to be heavier during morning (7-9 AM) and evening (5-7 PM) rush hours, especially near major towns like Sankari and Konganapuram. Key congestion-prone areas include town centers and intersections on Bhavani Main Road and SH 86. Weekends and local market days may see increased traffic from agricultural or commercial vehicles.

4. Road Quality and Infrastructure: The primary highways (SH 86 and Bhavani Main Road) are generally well-maintained, though rural segments may have narrower lanes and less visibility at night due to minimal street lighting. Watch for potholes and unpaved stretches in less-trafficked rural areas. Signage can sometimes be lacking, so route familiarity is a plus.

5. Alternative Routes for Emergencies: In case of road blockages or emergencies, consider:

- Detouring onto smaller local roads; however, these could involve slower speeds and less direct routes.
- Consulting local authorities for real-time advice on alternative routes.
- Using GPS navigation systems for live traffic updates to find efficient bypass paths.

6. Local Regulations: There are several regulations affecting the transport of hazardous materials, including limits on quantities, designated transport hours, and specific transportation permits. Ensure all documentation is in place and consult the Tamil Nadu transport authority for specific guidelines.

7. Historical Incidents: While not frequently reported, incidents involving heavy vehicles often relate to vehicle overturns or skidding in wet conditions, particularly on curves and inclines. Review historical data from local police logs for better insights into hazardous spots along this route.

8. Environmental Considerations: The route passes through areas with agricultural activity and small water bodies. Prevent fuel leaks and cargo spillage to mitigate environmental impact. Be cautious near sensitive zones and abide by environmental regulations.

9. Communication Coverage: Overall, mobile network coverage is generally reliable along the main highways and near towns. Expect possible dead zones in rural or forested areas, where alternative communication means like satellite phones or radios could be essential.

10. Emergency Response Times: Emergency response times can vary significantly:

- Urban areas like Sangagiri might have quicker response times due to proximity to facilities.
- Rural areas may experience delays, depending on distance to nearest service point. Collaborate with local emergency services for preemptive planning and ensure all personnel are aware of contact protocols.

11. Overall Risk Assessment Summary: The overall risk assessment along this route is moderate due to potential weather disruptions, variable road conditions, and occasional traffic congestion. Preparedness involves ensuring vehicle reliability, documenting and understanding local regulations, and remaining

vigilant of environmental and rural factors. Regular updates on weather and traffic conditions will help mitigate unforeseen challenges.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45847, 77.85140	30 KM/Hr
5	Turn	Medium	11.45898, 77.85135	30 KM/Hr
6	Turn	High	11.46303, 77.84945	15 KM/Hr
7	Turn	Medium	11.45571, 77.81702	30 KM/Hr
8	Turn	High	11.45615, 77.81677	15 KM/Hr
9	Blind Spot	Blind Spot	11.49395, 77.88537	10 KM/Hr
10	Turn	High	11.49465, 77.88488	15 KM/Hr
11	Turn	Medium	11.56096, 77.89931	30 KM/Hr
12	Turn	Medium	11.57035, 77.90223	30 KM/Hr
13	Turn	High	11.57041, 77.90221	15 KM/Hr
14	Turn	Medium	11.57649, 77.90610	30 KM/Hr
15	Turn	Medium	11.59214, 77.90656	30 KM/Hr
16	Turn	Medium	11.63833, 77.94818	30 KM/Hr
17	Turn	Medium	11.64171, 77.95184	30 KM/Hr
18	Turn	Medium	11.64243, 77.95198	30 KM/Hr
19	Turn	Medium	11.64721, 77.95118	30 KM/Hr
20	Turn	Medium	11.67870, 77.96846	30 KM/Hr
21	Turn	High	11.70227, 77.97864	15 KM/Hr
22	Turn	Medium	11.71715, 77.99969	30 KM/Hr
23	Turn	Medium	11.71921, 78.01133	30 KM/Hr
24	Turn	Medium	11.72737, 78.02248	30 KM/Hr
25	Turn	Medium	11.73775, 78.03465	30 KM/Hr
26	Turn	High	11.73651, 78.03878	15 KM/Hr
27	Turn	Medium	11.74189, 78.04660	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
28	Turn	Medium	11.74309, 78.04725	30 KM/Hr
29	Turn	Medium	11.78039, 78.05792	30 KM/Hr
30	Turn	Medium	11.86753, 78.08331	30 KM/Hr
31	Turn	Medium	11.86981, 78.08387	30 KM/Hr
32	Turn	High	11.87099, 78.08358	15 KM/Hr
33	Turn	High	11.87110, 78.08383	15 KM/Hr
34	Turn	Medium	11.87072, 78.08439	30 KM/Hr
35	Turn	Medium	11.87101, 78.08575	30 KM/Hr
36	Turn	High	11.87257, 78.08730	15 KM/Hr
37	Turn	Medium	11.86865, 78.12133	30 KM/Hr
38	Turn	Medium	11.86480, 78.12555	30 KM/Hr
39	Turn	High	11.85820, 78.13580	15 KM/Hr
40	Turn	Medium	11.86055, 78.13949	30 KM/Hr
41	Turn	Medium	11.86364, 78.14018	30 KM/Hr
42	Turn	Medium	11.86652, 78.13879	30 KM/Hr
43	Turn	Medium	11.88431, 78.14477	30 KM/Hr
44	Turn	Medium	11.88507, 78.14461	30 KM/Hr
45	Turn	Medium	11.90605, 78.15284	30 KM/Hr
46	Turn	High	11.90780, 78.15207	15 KM/Hr
47	Turn	Medium	11.91292, 78.15606	30 KM/Hr
48	Turn	High	11.92457, 78.17089	15 KM/Hr
49	Turn	Medium	11.92814, 78.18009	30 KM/Hr
50	Turn	High	11.92796, 78.18192	15 KM/Hr
51	Turn	Medium	11.92911, 78.18356	30 KM/Hr
52	Turn	Medium	11.93579, 78.18720	30 KM/Hr
53	Turn	Medium	11.95103, 78.20620	30 KM/Hr
54	Turn	Medium	11.95255, 78.20816	30 KM/Hr
55	Turn	Medium	11.95174, 78.21000	30 KM/Hr
56	Turn	Medium	11.95209, 78.21208	30 KM/Hr
57	Turn	High	11.95159, 78.21301	15 KM/Hr
58	Turn	High	11.95237, 78.21440	15 KM/Hr
59	Turn	High	11.95219, 78.21505	15 KM/Hr
60	Turn	Medium	11.95300, 78.21657	30 KM/Hr
61	Turn	Medium	11.95257, 78.22499	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
62	Turn	High	11.95180, 78.22589	15 KM/Hr
63	Turn	Medium	11.95527, 78.23224	30 KM/Hr
64	Turn	High	11.95860, 78.23412	15 KM/Hr
65	Turn	Medium	11.95900, 78.23399	30 KM/Hr
66	Turn	Medium	11.96157, 78.23446	30 KM/Hr
67	Turn	High	11.96662, 78.23292	15 KM/Hr
68	Blind Spot	Blind Spot	11.96688, 78.23203	10 KM/Hr
69	Turn	High	11.97329, 78.23713	15 KM/Hr
70	Turn	High	11.97751, 78.23549	15 KM/Hr
71	Turn	High	11.98245, 78.24303	15 KM/Hr
72	Turn	High	11.98209, 78.24350	15 KM/Hr
73	Turn	Medium	11.99357, 78.25820	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
10	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
12	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
13	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
14	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
15	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
16	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
17	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
18	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
19	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
20	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
21	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
24	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
26	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
27	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
28	clinic	Kamala Clinic	11.8723673, 78.0874195	30 km/h	Medium
29	hospital	Kamala Hospital	11.8722588, 78.0899189	30 km/h	Medium
30	hospital	GJB Hospital	11.8713998, 78.0980288	30 km/h	Medium
31	clinic	Pranav Clinic	11.8712846, 78.0998167	30 km/h	Medium
32	hospital	Bethel Hospital	11.8699472, 78.1388598	30 km/h	Medium
33	hospital	Grs Hospital	11.9824068, 78.2481763	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
11	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
23	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
25	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45898, 77.85135



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45571, 77.81702



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45615, 77.81677



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49395, 77.88537



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49465, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56096, 77.89931



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57035, 77.90223



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57041, 77.90221



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57649, 77.90610



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59214, 77.90656



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63833, 77.94818



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64171, 77.95184



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64243, 77.95198



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64721, 77.95118



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67870, 77.96846



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70227, 77.97864



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71715, 77.99969



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71921, 78.01133



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72737, 78.02248



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73775, 78.03465



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73651, 78.03878



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74189, 78.04660



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74309, 78.04725



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.78039, 78.05792



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86753, 78.08331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86981, 78.08387



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.87099, 78.08358



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.87110, 78.08383



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87072, 78.08439



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87101, 78.08575



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.87257, 78.08730



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95103, 78.20620



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.97329, 78.23713



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.97751, 78.23549



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.98245, 78.24303



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.98209, 78.24350



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.99357, 78.25820

